

MATHEMATICS AND STATISTICS

TIME: 3 HOURS

[MAX. MARKS: 80]

Note: (i) All questions are compulsory.

(ii) Figures to the right indicate full marks.

(iii) Graph of L.P.P should be drawn on graph paper only.

(iv) Use of logarithmic table is allowed.

(v) Answers to the questions of Section – I and Section –II should be written in the only one answer book.

(vi) Answer to every new question must be written on a new page.

SECTION -I

Question 1.

[12]

(A) Select and write the appropriate answer from the given alternatives in each of the following sub-equation:

(i) Which of the following represents direction cosines of the line: (2)

- (a) $0, \frac{1}{\sqrt{2}}, \frac{1}{2}$ (b) $0, \frac{-\sqrt{3}}{2}, \frac{1}{\sqrt{2}}$ (c) $0, \frac{-\sqrt{3}}{2}, \frac{1}{2}$ (d) $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$

(ii) $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $A(adjA) = KI$, then the value of 'K' is..... (2)

- (a) 2 (b) -2 (c) 10 (d) -10

(iii) The general solution of the trigonometric equation $\frac{1}{2}$ is..... (2)

- (a) $\theta = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ (b) $\theta = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$

- (c) $\theta = n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$ (d) $\theta = n\pi, n \in \mathbb{Z}$

(B) Attempt any THREE of the following:

(i) $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the points A, B, C respectively and $2\vec{a} + 3\vec{b} - 5\vec{c}$, then find the ratio in which the point C divides line segment AB. (2)

(ii) The Cartesian equation of a line is $\frac{x-6}{2}, \frac{y+4}{7}, \frac{z-5}{3}$, find its vector equation. (2)

(iii) Equation of a plane is $\vec{r} \cdot (3\hat{i} - 4\hat{j} + 12\hat{k}) = 8$. Find the length of the perpendicular from the origin to the plane. (2)

(iv) Find the acute angle between the lines whose direction ratios are 5, 12, -13 and 3, -4, 5 (2)

(v) Write the dual of the following statements: (2)

(1) $(p \vee q) \wedge T$

(2) Madhuri has curly hair and brown eyes.

Solution 1:

(A) (i) We have, the direction cosine of a line is l, m, n then,

$$l^2 + m^2 + n^2 = 1$$

$$0^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = 1$$

Hence, the correct option is (c), s

(ii) Given that,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\therefore (\text{Adj of } A) = KI$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = K \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} K & 0 \\ 0 & K \end{bmatrix}$$

$$\therefore K = -2$$

Hence, the correct option is (b).

(iii)

$$\tan^2 \theta = \tan \alpha$$

$$\theta = n\pi \pm \alpha$$

$$\tan^2 \theta = 1$$

$$\tan^2 \theta = \tan^2 \frac{\pi}{4}$$

$$\theta = n\pi \pm \frac{\pi}{4}$$

Hence, the correct option is (c).

(B) (i) Let the ratio be $\lambda : 1$

Position vector of point (C)

$$\vec{c} = \frac{\lambda \vec{a} + \vec{b}}{\lambda + 1}$$

Put the value of $\vec{c}$ in $2\vec{a} + 3\vec{b} - 5\vec{c} = 0$

$$2\vec{a} + 3\vec{b} - 5 \times \left(\frac{\lambda \vec{a} + \vec{b}}{\lambda + 1} \right) = 0$$

$$\Rightarrow \frac{(\lambda + 1)(2\vec{a} + 3\vec{b}) - 5\lambda \vec{a} - 5\vec{b}}{(\lambda + 1)} = 0$$

$$\Rightarrow 2\vec{a}\lambda + 3\vec{b}\lambda + 2\vec{a} + 3\vec{b} - 5\lambda \vec{a} - 5\vec{b} = 0$$

$$\Rightarrow 3\vec{b}\lambda - 3\vec{a}\lambda + 2\vec{a} - 2\vec{b} = 0$$

$$\Rightarrow 3\lambda(\vec{b} - \vec{a}) = 2\vec{b} - 2\vec{a}$$

$$\Rightarrow 3\lambda = \frac{2(\vec{b} - \vec{a})}{(\vec{b} - \vec{a})}$$

$$\Rightarrow \frac{\lambda}{1} = \frac{2}{3}$$

Hence, the point C divides line segment AB in ratio 2 : 3.

(ii) We have the equation of line,

$$\frac{x-6}{2} = \frac{y+4}{7} = \frac{z-5}{3}$$

Therefore, the line is passing through the point (6, -4, 5) and having direction ratio 2, 7, 3.

So,

$$\vec{a} = 6\hat{i} - 4\hat{j} + 5\hat{k}$$

and,

$$\vec{b} = 2\hat{i} + 7\hat{j} + 3\hat{k}$$

∴ The vector equation of line is $\vec{r} = \vec{a} + \lambda\vec{b}$

$$\Rightarrow \vec{r} = (6\hat{i} - 4\hat{j} + 5\hat{k}) + \lambda(2\hat{i} + 7\hat{j} + 3\hat{k})$$

(iii) Equation of plane is $\vec{r} \cdot (3\hat{i} - 4\hat{j} + 12\hat{k}) = 8$

Compare $\vec{r} \cdot \vec{n} = p$

Then,

$$\vec{n} = (3\hat{i} - 4\hat{j} + 12\hat{k}), p = 8$$

Therefore, length of the perpendicular from the origin to the plane is

$$\begin{aligned} d &= \frac{p}{|\vec{n}|} = \frac{8}{\sqrt{3^2 + (-4)^2 + 12^2}} \\ &= \frac{8}{\sqrt{9 + 16 + 144}} = \frac{8}{\sqrt{169}} \\ &= \frac{8}{13} \text{ unit.} \end{aligned}$$

(iv)

We have,

$$a_1 = 5, b_1 = 12, c_1 = -13$$

$$a_2 = 3, b_2 = -4, c_2 = 5$$

Acute angle,

$$\begin{aligned}\cos \theta &= \left| \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \right| \\ \Rightarrow \cos \theta &= \left| \frac{15 - 48 - 65}{\sqrt{25 + 144 + 169} \sqrt{9 + 16 + 25}} \right| \\ \Rightarrow \cos \theta &= \left| \frac{-98}{13\sqrt{2} \times 5 \times \sqrt{2}} \right| \\ \Rightarrow \cos \theta &= \left| \frac{-98}{13 \times 5 \times 2} \right| \\ \Rightarrow \cos \theta &= \left| \frac{-49}{65} \right| \\ \Rightarrow \cos \theta &= \frac{49}{65} \\ \therefore \theta &= \cos^{-1} \left(\frac{49}{65} \right)\end{aligned}$$

(v) Dual statements are

(1) $(p \wedge q) \vee T$

(2) Madhuri has curly hair or brown eyes.

Question 2

[14]

(A) Attempt any TWO of the following:

(i) If the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{2} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then find the value of k. (3)

(ii) Prove that three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar, if and only if, there exists a non-zero linear combination $x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$ (3)

(iii) Using truth table prove that, (3)

$$\sim p \wedge q$$

$$\equiv (p \vee q) \wedge \sim p$$

(B) Attempt any TWO of the following:

(i) In any $\triangle ABC$, with usual notations, prove that $b^2 = c^2 + a^2 - 2ca \cos B$. (4)

(ii) Show that the equation $x^2 - 6xy + 5y^2 + 10x - 14y + 9 = 0$ represents a pair of lines. Find the acute angle between them. Also find the point of intersection of the line. (4)

(iii) Express the following equations in the matrix form and solve them by method of reduction:

$$2x - y + z = 1, x + 2y + 3z = 8, 3x + y - 4z = 1 \quad (4)$$

Solution 2:

(i) The equation of the given lines is

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} \text{ and } \frac{x-3}{2} = \frac{y-k}{2} = \frac{z}{1}$$

$$\therefore x_1 = 1, y_1 = -1, z_1 = 1, x_2 = 3, y_2 = k, z_2 = 0$$

$$l_1 = 2, m_1 = 3, n_1 = 4, l_2 = 1, m_2 = 2, n_2 = 1$$

Since these lines intersect each other, we get

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 2 & k+1 & -1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 2(3-8) - (k+1)(2-4) - 1(4-3) = 0$$

$$\Rightarrow -10 + 2k + 2 - 1 = 0$$

$$\Rightarrow 2k = 9 \Rightarrow k = \frac{9}{2}$$

(ii) $\vec{a}, \vec{b}, \vec{c}$ be coplanar vectors. Then any one of them, say $\vec{a}$ will be the linear combination of $\vec{b}$ and $\vec{c}$.

$\therefore$ There exist scalars α and β such that,

$$\vec{a} = \alpha\vec{b} + \beta\vec{c}$$

$$\therefore (-1)\vec{a} + \alpha\vec{b} + \beta\vec{c} = 0$$

Where $x = -1, y = \alpha, z = \beta$ which are not all zero simultaneously then

$$x\vec{a} + y\vec{b} + z\vec{c} = 0$$

Conversely: let there exist scalars x, y, z not all zero such that,

$$x\vec{a} + y\vec{b} + z\vec{c} = 0$$

Let $x \neq 0$, then divides (i) by x , we get

$$x \left[\vec{a} + \frac{y}{x}\vec{b} + \frac{z}{x}\vec{c} \right] = 0$$

$$\vec{a} + \frac{y}{x}\vec{b} + \frac{z}{x}\vec{c} = 0$$

Hence, x, y, z are not zero scalars.

(iii) We have,

$$\sim p \wedge q \equiv (p \vee q) \wedge \sim p$$

Suppose $\sim p \wedge q = A(\text{say})$

and $(p \vee q) \wedge \sim p = B(\text{say})$

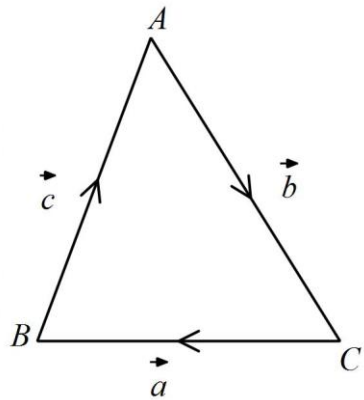
We have to show $A \equiv B$

p	q	$\sim p$	$p \vee q$	A	B
T	T	F	T	F	F
T	F	F	T	F	F
F	T	T	T	T	T
F	F	T	F	F	F

Since all truth values of column A and B are same or logically equivalent

Hence, $A \equiv B$ or $\sim p \wedge q \equiv (p \vee q) \wedge \sim p$

(B) (i) Let be ΔABC



In $\triangle ABC$,

$$\overrightarrow{BC} = \vec{a}, \overrightarrow{CA} = \vec{b}$$

$$\overrightarrow{BA} = \vec{c}$$

By law of sum of Δ 's

$$\vec{a} + \vec{b} = \vec{c}$$

$$\vec{b} = \vec{c} - \vec{a}$$

$$\vec{b} \cdot \vec{b} = \vec{b} \cdot (\vec{c} - \vec{a})$$

$$b^2 = (\vec{c} - \vec{a}) \cdot (\vec{c} - \vec{a})$$

$$b^2 = c^2 + a^2 - 2\vec{a} \cdot \vec{c}$$

$$\text{But, } \vec{a} \cdot \vec{c} = ac \cos B$$

$$\therefore b^2 = c^2 + a^2 - 2ca \cos B$$

(ii)

$$\text{Equation } x^2 - 6xy + 5y^2 + 10x - 14y + 9 = 0$$

$$\text{Compare with } ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

Then,

$$a = 1, h = -3, b = 5, g = 5, f = -7, c = 9$$

If equation represents a pair of line then,

$$\begin{aligned}
 abc + 2 fgh - af^2 - bg^2 - ch^2 &= 0 \\
 \therefore abc + 2 fgh - af^2 - bg^2 - ch^2 \\
 &= 1 \times 5 \times 9 + 2(-7) \times 5(-3) - 1(49) - 5 \times 25 - 9 \times 9 \\
 &= 45 + 210 - 49 - 125 - 81 \\
 &= 255 - 255 \\
 &= 0
 \end{aligned}$$

Hence, the given equation represents a pair of lines.

Acute angle,

$$\begin{aligned}
 \tan \theta &= \frac{2\sqrt{h^2 - ab}}{a + b} \\
 \tan \theta &= \frac{2\sqrt{9 - 5}}{1 + 5} = \frac{2 \times 2}{6} \\
 \theta &= \tan^{-1}\left(\frac{2}{3}\right)
 \end{aligned}$$

Point of intersection

$$\begin{aligned}
 &\left(\frac{hf - bg}{ab - h^2}, \frac{hg - af}{ab - h^2} \right) \\
 &= \left(\frac{(-3) \times (-7) - 5 \times 5}{5 - 9}, \frac{(-3) \times 5 - 1 \times (-7)}{5 - 9} \right) \\
 &= \left(\frac{21 - 25}{-4}, \frac{-15 + 7}{-4} \right) \\
 &= \left(\frac{-4}{-4}, \frac{-8}{-4} \right)
 \end{aligned}$$

$\therefore$ intersection point is (1, 2)

(iii)

The given equation can be written in the matrix form as:

$$\begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 1 \end{bmatrix}$$

Applying $R_1 \leftrightarrow R_2$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \\ 1 \end{bmatrix}$$

Applying $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -5 \\ 0 & -5 & -13 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -15 \\ -23 \end{bmatrix}$$

Applying $R_3 \rightarrow R_3 - R_2$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -5 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -15 \\ -8 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x+2y+3z \\ 0-5y-5z \\ 0+0-8z \end{bmatrix} = \begin{bmatrix} 8 \\ -15 \\ -8 \end{bmatrix}$$

By equality of matrices,

$$x+2y+3z=8 \text{----- (i)}$$

$$-5y-5z=-15 \text{----- (ii)}$$

$$-8z=-8 \text{----- (iii)}$$

From equation (iii), $z = 1$

Putting $z = 1$ in equation (ii), we get

$$-5y = -15 + 5$$

$$-5y = -10$$

$$y = 2$$

Putting the value of y and z in equation (i), we get

$$x + 2(2) + 3(1) = 8$$

$$x = 8 - 7 = 1$$

Hence, $x = 1, y = 2$ and $z = 1$ is the required solution.

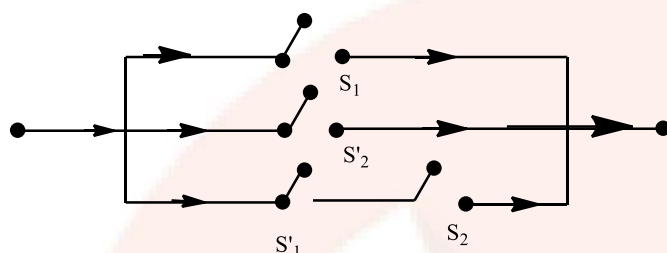
Question 3

[14]

(A) Attempt any TWO of the following:

(i) Prove that a homogeneous equation of degree two in x and y (i.e. $ax^2 + 2hxy + by^2 = 0$), Represents a pair of lines through the origin if $h^2 - ab \geq 0$. (3)

(ii) Find the symbolic form of the following switching circuit, construct its switching table and interpret it. (3)



(iii) If A, B, C, D are $(1, 1, 1), (2, 1, 3), (3, 2, 2)$ and $(3, 3, 4)$ respectively. Then find the volume of the parallelepiped with AB, AC and AD as concurrent edges. (3)

(B) Attempt any TWO of the following:

(i) Find the equation of the plane passing through the line of intersection of planes $2x - y + z = 3$, $4x - 3y + 5z + 9 = 0$ and parallel to the line $\frac{x+1}{2} = \frac{y+3}{4} = \frac{z-3}{5}$ (4)

(ii) Minimize: $Z = 6x + 4y$ (4)

Subject to:

$$3x + 2y \geq 12,$$

$$x + y \geq 5,$$

$$0 \leq x \leq 4,$$

$$0 \leq y \leq 4$$

(iii) Show that: $\cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) = \cos^{-1}\left(\frac{33}{65}\right)$. (4)

Solution 3:

(A)

(i) Let us consider a homogeneous equation of degree two in x and y .

Then,

$$ax^2 + 2hxy + by^2 = 0 \text{-----(1)}$$

Where, either a or b or h is non-zero.

We consider two case:

Case 1

If $b = 0$, then equation becomes

$$\begin{aligned} ax^2 + 2hxy &= 0 \\ x(ax + 2hy) &= 0 \end{aligned}$$

Which is a joint equation of lines $x = 0$ and $ax + 2hy = 0$.

Thus, the lines pass through the origin.

Case 2:

If $b \neq 0$

Multiplying both sides in equation (1) by b, we get

$$\begin{aligned} abx^2 + 2hbxy + b^2y^2 &= 0 \\ b^2y^2 + 2hbxy &= -abx^2 \\ b^2y^2 + 2hbxy + h^2x^2 &= -abx^2 + h^2x^2 \\ (by + hx)^2 &= \left[\left(\sqrt{h^2 - ab} \right) x \right]^2 \\ (by + hx)^2 - \left[\left(\sqrt{h^2 - ab} \right) x \right]^2 &= 0 \\ \left[(by + hx) + \left(\sqrt{h^2 - ab} \right) x \right] \left[(by + hx) - \left(\sqrt{h^2 - ab} \right) x \right] &= 0 \end{aligned}$$

Now, joint equation of two lines,

$$(by + hx) + \left(\sqrt{h^2 - ab} \right) x = 0 \text{ and } (by + hx) - \left(\sqrt{h^2 - ab} \right) x = 0$$

$$\left(h + \sqrt{h^2 - ab} \right) x + by = 0 \text{ and } \left(h - \sqrt{h^2 - ab} \right) x + by = 0$$

These lines pass through the origin when $h^2 - ab \geq 0$.

Hence, the equation $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines passing through the origin.

(ii) Symbolic form = $(S_1 + S_2') + (S_1 \cdot S_2)$

Switching table

S_1	S_2	S_1'	S_2'	$S_1 + S_2'$	$S_1' S_2$	$S_1 + S_2' + S_1' S_2$
T	T	F	F	T	F	T
T	F	F	T	T	F	T
F	T	T	F	F	T	T
F	F	T	T	T	F	T

All the entries the last column of the above truth table are T.

$\therefore S_1 + S_2' + S_1' S_2$ is tautology.

(iii) Given that,

A, B, C and D are (1, 1, 1), (2, 1, 3), (3, 2, 2) and (3, 3, 4) respectively. We need to find the volume of the parallelepiped with AB, AC, and AD as the concurrent edges.

The volume of the parallelepiped whose edges are $\vec{a}$, $\vec{b}$ and $\vec{c}$ is $[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c})$

$$\vec{AB} = (2-1)\hat{i} + (1-1)\hat{j} + (3-1)\hat{k}$$

$$= \hat{i} + 2\hat{k}$$

$$\vec{AC} = (3-1)\hat{i} + (2-1)\hat{j} + (2-1)\hat{k}$$

$$= 2\hat{i} + \hat{j} + \hat{k}$$

$$\vec{AD} = (3-1)\hat{i} + (3-1)\hat{j} + (4-1)\hat{k}$$

$$= 2\hat{i} + 2\hat{j} + 3\hat{k}$$

$$[\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 2 & 2 & 3 \end{vmatrix}$$

$$= 1(3-2) - 0 + 2(4-2)$$

$$= 1 + 4$$

$$= 5 \text{ cubic units}$$

(B) (i)

Equation of the plane passing through the line of intersection of planes $(2x - y + z) - 3 = 0$ and $4x - 3y + 5z + 9 = 0$ is

$$(2x - y + z - 3) + \lambda(4x - 3y + 5z + 9) = 0 \text{ --- (1)}$$

$$(2 + 4\lambda)x - (1 + 3\lambda)y + (1 + 5\lambda)z - 3 + 9\lambda = 0$$

Since this plane is parallel to the line $\frac{x+1}{2} = \frac{y+3}{4} = \frac{z-3}{5}$ whose direction ratio are 2,4,5

∴ The normal to the plane is perpendicular to this line.

$$\therefore (2+4\lambda)2 + [-(1+3\lambda)]4 + (1+5\lambda)5 = 0$$

$$\Rightarrow 4 + 8\lambda - 4 - 12\lambda + 5 + 25\lambda = 0$$

$$\Rightarrow 5 + 21\lambda = 0$$

$$\Rightarrow \lambda = \frac{-5}{21}$$

Substituting $\lambda = \frac{-5}{21}$ in equation(i), we get

$$(2x - y - z - 3) + \left(\frac{-5}{21}\right)(4x - 3y + 5z + 9) = 0$$

$$42x - 21y + 21z - 63 - 20x + 15y - 25z - 45 = 0$$

$$22x - 6y - 4z - 108 = 0$$

$$2(11x - 3y - 2z - 54) = 0$$

$$11x - 3y - 2z - 54 = 0$$

(ii)

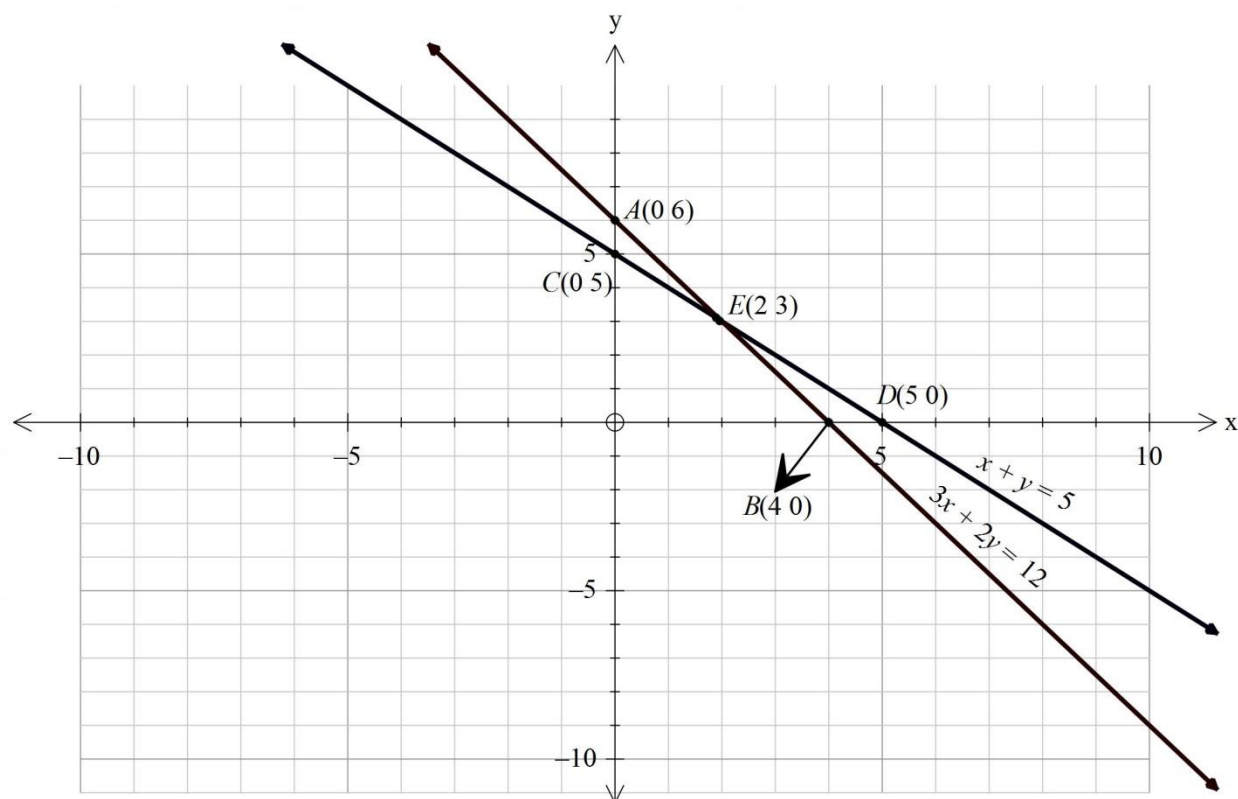
First we draw the line AB and CD whose equation are:

$$3x + 2y = 12$$

	A	B
x	0	4
Y	6	0

$$x + y = 5$$

	C	D
x	0	5
Y	5	0



The feasible region is XDEAY which is shaded in the figure.

The vertices of the feasible region are $A(0, 6)$ and $D(5, 0)$.

E is the point of intersection of the line. Now solving equation (i) and (ii) we get intersection point $E(2, 3)$. Origin does not satisfy given condition $(0, 0)$ does not present in the shaded area.

Calculation of minimum value:

Corner points	$Z = 6x + 4y$
At $A(0, 6)$	$Z = 6 \times 0 + 4 \times 6 = 24$
At $E(2, 3)$	$Z = 6 \times 2 + 4 \times 3 = 24$
At $D(5, 0)$	$Z = 6 \times 5 + 4 \times 0 = 30$

Since Z has minimum value 24 at two consecutive vertices A and E of the feasible region, Z has minimum value 24 at every point of segment AE .

Hence, there are infinite number of optimal solutions.

(iii)

Given that,

$$\cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) = \cos^{-1}\left(\frac{33}{65}\right)$$

$$\therefore \cos^{-1} x + \cos^{-1} y = \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right)$$

Taking L.H.S,

$$\begin{aligned}\cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) &= \cos^{-1}\left(\frac{4}{5} \times \frac{12}{13} - \sqrt{1-\frac{16}{25}}\sqrt{1-\frac{144}{169}}\right) \\ &= \cos^{-1}\left(\frac{48}{65} - \sqrt{\frac{9}{25}} \times \sqrt{\frac{25}{169}}\right) \\ &= \cos^{-1}\left(\frac{48}{65} - \frac{15}{65}\right) \\ &= \cos^{-1}\left(\frac{33}{65}\right)\end{aligned}$$

Hence, LHS= RHS proved.

SECTION-II

Question 4

[12]

(A) Select and write the appropriate answer from the given alternatives in each of the following sub-questions:

(i) If $y = 1 - \cos \theta$, $x = 1 - \sin \theta$ then $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$ is (2)

- (a) -1 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}$

(ii) The integrating factor of linear differential equation $\frac{dy}{dx} + y \sec x = \tan x$ is (2)

- (a) $\sec x - \tan x$ (b) $\sec x \cdot \tan x$ (c) $\sec x + \tan x$ (d) $\sec x \cdot \cot x$

(iii) The equation of tangent to the curve $y = 3x^2 - x + 1$ at the point (1, 3) is (2)

(a) $y = 5x + 2$ (b) $y = 5x - 2$ (c) $y = \frac{1}{5}x + 2$ (d) $y = \frac{1}{5}x - 2$

(B) Attempt any THREE of the following:

(i) Examine the continuity of the function (2)

$$f(x) = \sin x - \cos x, \text{ for } x \neq 0$$

$$= -1, \text{ for } x = 0$$

At the point $x = 0$

(ii) Verify Rolle's theorem for the function $f(x) = x^2 - 5x + 9$ on $[1, 4]$ (2)

(iii) Evaluate: $\int \sec^n x \tan x dx$ (2)

(iv) The probability mass function (p.m.f.) of X is given below: (2)

$X = x$	1	2	3
$P(X = x)$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{2}{5}$

Find $y = 1$

(v) Given that $X \sim B(n = 10, p)$ if $E(X) = 8$, find the value of p . (2)

Solution 4:

(A) (i) Let,

$$y = 1 - \cos \theta \Rightarrow \frac{dy}{d\theta} = \sin \theta$$

$$x = 1 - \sin \theta \Rightarrow \frac{dx}{d\theta} = -\cos \theta$$

$$\therefore \frac{dy}{dx} = -\tan \theta$$

$$\theta = \frac{\pi}{4}$$

$$\left[\frac{dy}{dx} \right]_{\theta = \frac{\pi}{4}} = -\tan \frac{\pi}{4} = -1$$

Hence, the correct option is (a).

(ii) Given that,

$$\frac{dy}{dx} + y \sec x = \tan x$$

Here,

$$P = \sec x$$

$$\therefore I.F = e^{\int P dx}$$

$$= e^{\int \sec x dx}$$

$$= e^{\log(\sec x + \tan x)}$$

$$= \sec x + \tan x$$

Hence, the correct option is (c).

(iii) We have,

$$y = 3x^2 - x + 1$$

$$\therefore \frac{dy}{dx} = 6x - 1$$

$$\left(\frac{dy}{dx} \right)_{(1,3)} = 6 - 1 = 5$$

$$\therefore m = 5$$

Equation of tangent

$$y = mx + c$$

$$y = 5x + c$$

Put (1, 3) in equation (i),

$$3 = 5 \times 1 + c$$

$$c = -2$$

Equation of tangent

$$y = 5x - 2$$

Hence, the correct option is (b).

(B) (i) Let,

$$L.H.L = \lim_{x \rightarrow 0^-} (\sin x - \cos x)$$

$$= \lim_{x \rightarrow 0^-} f(x)$$

$$= 0 - 1$$

$$= -1$$

$$R.H.L = \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{x \rightarrow 0^+} (\sin x - \cos x) = 1$$

$$\lim_{x \rightarrow 0} f(x) = -1 \text{ (given)}$$

$$L.H.L = R.H.L = \lim_{x \rightarrow 0} f(x)$$

So, function is continuous at $x = 0$

(ii) Let us consider the given function $f(x) = x^2 - 5x + 9$ is a polynomial.

Therefore, it is continuous and differentiable in its domain $x \in R$. Hence it is continuous in the interval $[1, 4]$ and differentiable in the interval $(1, 4)$

$$f(1) = 1 - 5 + 9 = 5$$

$$f(4) = 16 - 20 + 9 = 5$$

$$f(1) = f(4)$$

$\therefore$ Conditions of Roll's theorem are satisfied.

$$f'(x) = 2x - 5$$

$$\Rightarrow f'(c) = 2c - 5$$

$$\Rightarrow f''(c) = 0$$

$$\Rightarrow 2c - 5 = 0, c = \frac{5}{2} \in [1, 4]$$

$$f'(c) = 0 \text{ at } c = \frac{5}{2}$$

(iii) Given that,

$$I = \int \sec^n x \tan x dx$$

$$I = \int \sec^{n-1} x \sec x \tan x dx$$

Let,

$$\sec x = t \Rightarrow \sec x \tan x dx = dt$$

$$\therefore I = \int t^{n-1} dt$$

$$= \frac{t^n}{n} + c$$

$$= \frac{\sec^n x}{n} + c$$

(iv) We have,

$$E(X^2) = \sum X^2 P(X)$$

$$E(X^2) = 1^2 \times \frac{1}{5} + 2^2 \times \frac{2}{5} + 3^2 \times \frac{2}{5}$$

$$= \frac{1}{5} + \frac{8}{5} + \frac{18}{5} = \frac{27}{5}$$

(v) Given that,

$$X \sim B(n=10, P)$$

$$E(X) = 8$$

$$E(X) = nP$$

$$\therefore nP = 8$$

$$10P = 8$$

$$P = 0.8$$

Hence, the value of p is 0.8.

Question 5.

[14]

(A) Attempt any TWO of the following:

(i) If $y = f(u)$ is a differentiable function of u and $u = g(x)$ is a differentiable function of x , then prove that $y = f[g(x)]$ is a differentiable function of x and $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$. **(3)**

(ii) Obtain the differential equation by eliminating arbitrary constants A, B from the equation-

$$y = A \cos(\log x) + B \sin(\log x) \quad (3)$$

(iii) Evaluate: $\int \frac{x^2}{(x^2 + 2)(2x^2 + 1)} dx$ **(3)**

(B) Attempt any TWO of the following:

(i) An open box is to be made out of a piece of a square cardboard of sides 18 cms. by cutting off equal squares from the corners and turning up the sides. Find the maximum volume of the box. (4)

(ii) Prove that:

$$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx \quad (4)$$

(iii) If the function $f(x)$ is continuous in the interval $[-2, 2]$, find the values of a and b where

$$\begin{aligned} f(x) &= \frac{\sin ax}{x} - 2, \text{ for } -2 \leq x < 0 \\ &= 2x+1, \text{ for } 0 \leq x \leq 1 \\ &= 2b\sqrt{x^2+3}-1, \text{ for } 1 < x \leq 2 \end{aligned} \quad (4)$$

Solution 5:

(A) (i) We have,

Let $\delta x, \delta y$ and δu be a small change in x, y and u respectively.

As, $\delta x \rightarrow 0, \delta y \rightarrow 0, \delta u \rightarrow 0$

As u is differentiable function, it is continuous.

Now, consider the increment ratio $\frac{\delta y}{\delta x}$

So, we have,

$$\frac{\delta y}{\delta x} = \frac{\delta y}{\delta u} \times \frac{\delta u}{\delta x}$$

Taking limit as $\delta x \rightarrow 0$ on both sides

$$\begin{aligned} \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} &= \lim_{\delta x \rightarrow 0} \left(\frac{\delta y}{\delta u} \times \frac{\delta u}{\delta x} \right) \\ \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} &= \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta u} \times \lim_{\delta x \rightarrow 0} \frac{\delta u}{\delta x} \\ \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} &= \lim_{\delta u \rightarrow 0} \frac{\delta y}{\delta u} \times \lim_{\delta x \rightarrow 0} \frac{\delta u}{\delta x} \text{----- (1)} \end{aligned}$$

Since, y is a differentiable function of u , $\lim_{\delta u \rightarrow 0} \left(\frac{\delta y}{\delta u} \right)$ exists and $\lim_{\delta x \rightarrow 0} \left(\frac{\delta u}{\delta x} \right)$ exists as

u is a differentiable function of x .

RHS of equation (1) exists.

Now,

$$= \lim_{\delta u \rightarrow 0} \frac{\delta y}{\delta u} = \frac{dy}{du} \text{ and } \lim_{\delta x \rightarrow 0} \frac{\delta u}{\delta x} = \frac{du}{dx}$$

$$\therefore \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta u} = \frac{dy}{du} \times \frac{du}{dx}$$

So, RHS exists.

Therefore, LHS also exists and

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Hence, proved.

(ii) We have,

$$y = A \cos(\log x) + B \sin(\log x)$$

Differentiating w.r.to x , we get

$$\frac{dy}{dx} = -A \sin(\log x) \times \frac{1}{x} + B \cos(\log x) \times \frac{1}{x}$$

$$x \frac{dy}{dx} = -A \sin(\log x) + B \cos(\log x)$$

Again, differentiating, we get

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = -A \cos(\log x) \times \frac{1}{x} - B \sin(\log x) \times \frac{1}{x}$$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = -[A \cos(\log x) + B \sin(\log x)]$$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = -y$$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$$

(iii) Given that,

$$I = \int \frac{x^2}{(x^2 + 2)(2x^2 + 1)} dx$$

Let,

$$\frac{x^2}{(x^2 + 2)(2x^2 + 1)} = \frac{A}{x^2 + 2} + \frac{B}{2x^2 + 1}$$

Put $x^2 = y$

$$\frac{y}{(y + 2)(2y + 1)} = \frac{A}{y + 2} + \frac{B}{2y + 1}$$

$$y = A(2y + 1) + B(y + 2)$$

Put $y + 2 \Rightarrow y = -2$

$$-2 = A(-3) + 0$$

$$A = \frac{2}{3}$$

Put $2y + 1 = 0 \Rightarrow y = -\frac{1}{2}$

$$-\frac{1}{2} = 0 + B\left(-\frac{1}{2} + 2\right)$$

$$-\frac{1}{2} = B\left(\frac{3}{2}\right)$$

$$\therefore B = -\frac{1}{3}$$

$$\therefore \frac{y}{(y+2)(2y+1)} = \frac{2}{3(y+2)} - \frac{1}{3(2y+1)}$$

Then,

$$\frac{x^2}{(x^2+2)(2x^2+1)} = \frac{2}{3(y+2)} - \frac{1}{3(2y+1)}$$

$$\int \frac{x^2}{(x^2+2)(2x^2+1)} dx = \int \frac{2}{3(y+2)} dx - \int \frac{dx}{3(2y+1)}$$

$$= \frac{2}{3} \int \frac{dx}{x^2 + (\sqrt{2})^2} - \frac{1}{3} \int \frac{dx}{2x^2 + 1}$$

$$= \frac{2}{3} \times \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} - \frac{1}{3 \times 2} \int \frac{dx}{x^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$$

$$= \frac{\sqrt{2}}{3} \tan^{-1} \frac{x}{\sqrt{2}} - \frac{1}{6} \times \frac{1}{1/\sqrt{2}} \tan^{-1} \frac{x}{1/\sqrt{2}} + c$$

$$= \frac{\sqrt{2}}{3} \tan^{-1} \frac{x}{\sqrt{2}} - \frac{1}{3\sqrt{2}} \tan^{-1} \sqrt{2}x + c$$

(B) (i) Let each side of the square cut off from each corner be x cm.

Then volume of box $V = (18-2x)(18-2x)x$

$$V = (18-2x)^2 x$$

$$V = 4x^3 + 324x - 72x^2 \text{ -----(i)}$$

Differentiating w.r.to x, we get

$$\frac{dV}{dx} = 12x^2 + 324 - 144x$$

$$\frac{dV}{dx} = 12(x^2 - 12x + 27) \text{ -----(ii)}$$

For maximum volume,

$$\frac{dV}{dx} = 0$$

$$\Rightarrow 12(x^2 - 12x + 27) = 0$$

$$\Rightarrow x^2 - 9x - 3x + 27 = 0$$

$$\Rightarrow (x - 9)(x - 3) = 0$$

$$\Rightarrow x = 9, 3$$

Again differentiating, we get

$$\frac{d^2V}{dx^2} = 2x - 12 \text{-----(iii)}$$

at $x = 9$

$$\frac{d^2V}{dx^2} = +ve$$

$\therefore V$ is minimum at $x = 9$

at $x = 3$

$$\frac{d^2V}{dx^2} = -ve$$

$\therefore V$ is maximum at $x = 3$

$$\begin{aligned} \therefore \text{Maximum volume } V &= (18 - 6)(18 - 6) \times 3 \\ &= 12 \times 12 \times 3 = 432 \text{ cm}^3 \end{aligned}$$

(ii) Given that,

$$\begin{aligned} R.H.S &= \int_0^a f(x)dx + \int_0^a f(2a - x)dx \\ &= I_1 + I_2 \end{aligned}$$

For,

$$I_2 = \int_0^a f(2a - x)dx$$

Put,

$$2a - x = t$$

$$\therefore dx = -dt$$

$$\text{when } x = 0 \text{ then } t = 2a$$

$$\text{when } x = a \text{ then } t = a$$

$$\begin{aligned} \int_0^a (2a - x)dx &= \int_{2a}^a f(t)(-dt) \\ &= -\int_{2a}^a f(t)dt \quad \left(\because \int_a^b f(x)dx = -\int_b^a f(x)dx \right) \end{aligned}$$

$$\therefore \int_0^a (2a - x)dx = \int_a^{2a} f(x)dx$$

$$\begin{aligned} \therefore I_1 + I_2 &= \int_0^a f(x)dx + \int_a^{2a} f(x)dx \\ &= \int_0^{2a} f(x)dx \end{aligned}$$

Hence, RHS = LHS proved.

(iii) We have,

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} \frac{\sin ax}{x} - 2 \\ &= \lim_{x \rightarrow 0^-} \left(a \frac{\sin ax}{ax} - 2 \right) \\ &= a \times 1 - 2 \quad \left(\because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right) \\ &= a - 2 \\ \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} (2x + 1) \\ &= 1 \end{aligned}$$

Function is continuous at $x = 0$

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^+} f(x) \\ a - 2 &= 1 \\ a &= 3 \end{aligned}$$

Again,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (2x + 1) \\ = 3$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 2b\sqrt{x^2 + 3} - 1 \\ = 2b\sqrt{4} - 1 = 4b - 1$$

Function is continuous at $x = 1$

Then,

$$4b - 1 = 3$$

$$4b = 4 \Rightarrow b = 1$$

Question 6.

[14]

(A) Attempt any TWO of the following:

(i) Solve the differential equation: $\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$ **(3)**

(ii) A fair coin is tossed 8 times. Find the probability that it shows heads at least once. **(3)**

(iii) If $x^p y^q = (x + y)^{p+q}$ then prove that $\frac{dy}{dx} = \frac{y}{x}$ **(3)**

(B) Attempt any TWO of the following:

(i) Find the area of the sector of a circle bounded by the circle $x^2 + y^2 = 16$ and the line $y = x$ in the first quadrant. **(4)**

(ii) Prove that: **(4)**

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + c$$

(iii) A random variable X has the following probability distribution:

(4)

$X = x$	0	1	2	3	4	5	6
$P(X = x)$	K	3k	5k	7k	9k	11k	13k

(a) Find k

(b) Find $P(0 < X < 4)$

(c) Obtain cumulative distribution function (c.d.f.) of X.

Solution 6:

(A) (i) Given that,

$$\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$$

Putting $y = vx \Rightarrow \frac{dy}{dv} = v + x \frac{dv}{dx}$

$$\therefore v = \frac{y}{x}$$

$\therefore$ Equation (i) becomes,

$$v + x \frac{dv}{dx} = \frac{vx + \sqrt{x^2 + v^2 x^2}}{x}$$

$$\therefore v + x \frac{dv}{dx} = v + \sqrt{1 + v^2}$$

$$\Rightarrow x \frac{dv}{dx} = \sqrt{1 + v^2}$$

$$\Rightarrow \frac{1}{\sqrt{1 + v^2}} dv = \frac{1}{x} dx$$

On integration, we get

$$\int \frac{1}{\sqrt{1 + v^2}} dv = \int \frac{1}{x} dx$$

$$\log |v + \sqrt{1 + v^2}| = \log |x| + \log c$$

$$\log \left| \frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} \right| = \log |cx|$$

$$\frac{y + \sqrt{x^2 + y^2}}{x} = cx$$

$$y + \sqrt{x^2 + y^2} = cx^2$$

Hence, this is the general solution.

(ii) P = come in head on a fair coin

$$p = \frac{1}{2}, q = 1 - \frac{1}{2} = \frac{1}{2}$$

If head don't come any time

$$\begin{aligned} &= {}^8C_0 p^0 q^8 \\ &= {}^8C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^8 \\ &= 1 \times \frac{1}{2^8} = \frac{1}{256} \end{aligned}$$

$$\therefore \text{show heads at least once } 1 - \frac{1}{256} = \frac{255}{256}$$

(iii) Given that,

$$x^p y^q = (x + y)^{p+q}$$

Take log on both sides, we get

$$p \log x + q \log y = (p + q) \log(x + y)$$

Differentiate w.r.to x

$$\frac{p}{x} + \frac{q}{y} \frac{dy}{dx} = \frac{p+q}{x+y} \left(1 + \frac{dy}{dx}\right)$$

$$\frac{dy}{dx} \left(\frac{q}{y} - \frac{p+q}{x+y} \right) = \frac{p+q}{x+y} - \frac{p}{x}$$

$$\frac{dy}{dx} \left(\frac{qx - py}{y(x+y)} \right) = \frac{qx - py}{x(x+y)}$$

$$\frac{dy}{dx} = \frac{y}{x}$$

(B) (i) Centre of circle is (0,0)

Radius of circle = 4

$$x^2 + y^2 = 16 \text{ -----(i)}$$

$$y = x \text{ -----(ii)}$$

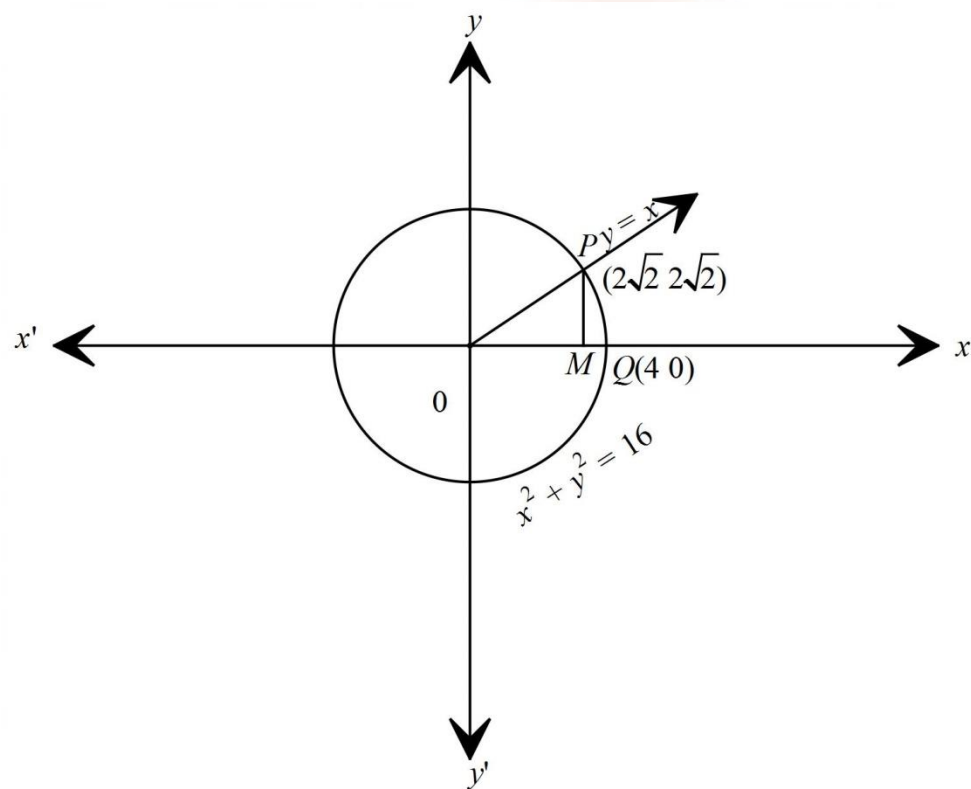
By equation (i) and (ii), we get

$$x^2 + y^2 = 16$$

$$\Rightarrow x^2 = 8$$

$$\therefore x = 2\sqrt{2}, y = 2\sqrt{2}$$

Shaded area = Area of OMPO + Area of PMQP



$$= \int_0^{2\sqrt{2}} y dx + \int_{2\sqrt{2}}^4 y dx$$

By line by circle

$$\begin{aligned}
 &= \int_0^{2\sqrt{2}} x dx + \int_{2\sqrt{2}}^4 \sqrt{16-x^2} dx \\
 &= \left[\frac{x^2}{2} \right]_0^{2\sqrt{2}} + \left[\frac{x}{2} \sqrt{16-x^2} + \frac{16}{2} \sin^{-1} \frac{x}{4} \right]_{2\sqrt{2}}^4 \\
 &= \left(\frac{8}{2} - 0 \right) + [0 + 8 \sin^{-1} 1] - \left[\frac{2\sqrt{2}}{2} \sqrt{16-8} + 8 \sin^{-1} \frac{1}{\sqrt{2}} \right] \\
 &= 4 + \left[8 \sin^{-1} 1 - \sqrt{2} \times 2\sqrt{2} - 8 \sin^{-1} \frac{1}{\sqrt{2}} \right] \\
 &= 4 + \left(8 \times \frac{\pi}{2} - 4 - 8 \times \frac{\pi}{4} \right) \\
 &= 4 + 4\pi - 4 - 2\pi \\
 &= 2\pi \text{ square unit.}
 \end{aligned}$$

(ii) Given that,

$$\begin{aligned}
 &\int \sqrt{x^2 - a^2} dx \\
 &= \sqrt{x^2 - a^2} (x) - \int \frac{2x}{2\sqrt{x^2 - a^2}} x dx \\
 &= \sqrt{x^2 - a^2} (x) - \int \frac{x^2}{\sqrt{x^2 - a^2}} dx \\
 &= \sqrt{x^2 - a^2} (x) - \int \frac{x^2 - a^2 + a^2}{\sqrt{x^2 - a^2}} dx \\
 &= \sqrt{x^2 - a^2} (x) - \int \frac{x^2 - a^2}{\sqrt{x^2 - a^2}} dx + a^2 \int \frac{1}{\sqrt{x^2 - a^2}} dx \\
 &= \sqrt{x^2 - a^2} (x) - \int \sqrt{x^2 - a^2} dx + a^2 \int \frac{1}{\sqrt{x^2 - a^2}} dx \\
 &\Rightarrow 2 \int \sqrt{x^2 - a^2} dx = \sqrt{x^2 - a^2} (x) + a^2 \int \frac{1}{\sqrt{x^2 - a^2}} dx \\
 &\Rightarrow \int \sqrt{x^2 - a^2} dx = \frac{1}{2} \sqrt{x^2 - a^2} (x) + \frac{1}{2} a^2 \log \left| x + \sqrt{x^2 - a^2} \right| + c
 \end{aligned}$$

Where c is constant term.

(iii) (a)

$$k + 3k + 5k + 7k + 9k + 11k + 13k = 1$$

$$\therefore 49k = 1$$

$$\therefore k = \frac{1}{49}$$

(b)

$$P(0 < x < 4) = P(1) + P(2) + P(3)$$

$$= \frac{3}{49} + \frac{10}{49} + \frac{21}{49} = \frac{34}{49}$$

(c)

$$c.d.f = (2x+1)K, 0 \leq x$$